Paper Review • Embodied AI / World Action Models

SWAM: Cross-Modally Aligned WAM Scaling

SWAM: Scaling World Action Models with Cross-Modal Alignment

Yu Zhou*, Xinyuan Xia*, Xuankun Yang*, Yinghao Yang*, … , Cewu Lu, Yanwei Li †
Shanghai Jiao Tong Univ. • Shanghai Innovation Institute • Noematrix • Westlake Univ. • Tongji Univ.

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TL;DR

Video-pretrained L–V alignment is key to WAM generalization

Using ~**47k hours** of multi-source data (general video, egocentric video, UMI handheld-gripper demos, robot demos), SWAM systematically studies how world action models can preserve language–video alignment while learning actions. The resulting recipe: **detailed subtask captions + UMI action labels + 1:1 Ego/UMI mix + action branch reading language only via video**, clearly outperforming existing VLAs and WAMs in unseen scenes.

47k h

Multi-Source Pretraining

4 types: General / Ego / UMI / Robot

61.56%

RoboTwin Hard (C2R) SR

12.86 pt above strongest baseline OpenWAM

88.10%

RoboTwin Easy (C2C) SR

On par with OpenWAM (89.30%)

40.00%

Real-Robot Hard SR

Avg. over 3 tasks; OpenWAM: 23.33%

Motivation

Action learning may erode video-pretrained L–V alignment

What Is a WAM?

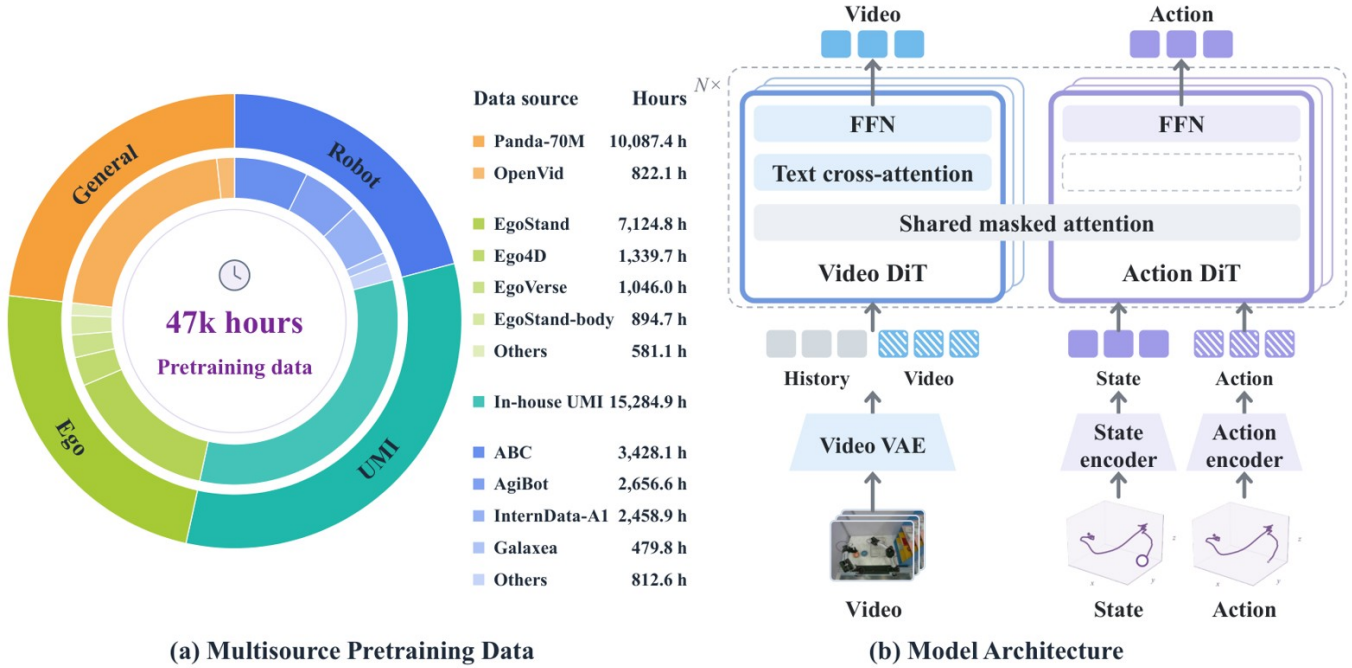
World Action Model: language-conditioned joint prediction of future video and actions, bringing the visual/dynamics priors of video models to robot control.

Blind Spot of Prior Work

Motus, LingBot-VA, Fast-WAM, OpenWAM, etc. mainly improve architectures and training, assuming the language–video alignment from video pretraining is a "given foundation".

Our Question

With action supervision, how can language–video priors be preserved and language–video–action correspondence built? What role does each data source and supervision type play?



(a) Multisource Pretraining Data (b) Model Architecture

Fig. 2: (a) 47k h multi-source PT data; (b) video/action shared masked attn.

Framework

3 alignment levels + scaling: 4 studies, 4 actionable lessons

01

Lang.–Video Align.

How to write captions?

Richer detail, scene/lighting, subtask-level alignment → Hard SR 47.20 → 55.04

02

L–V–A Alignment

Do human-demo actions help?

UMI action-label coverage 0% → 100%: Hard SR 46.72 → 51.28

03

Co-Training

What mix? How to connect?

Ego:UMI = 1:1 is best; action branch reading language only via video: 57.36

04

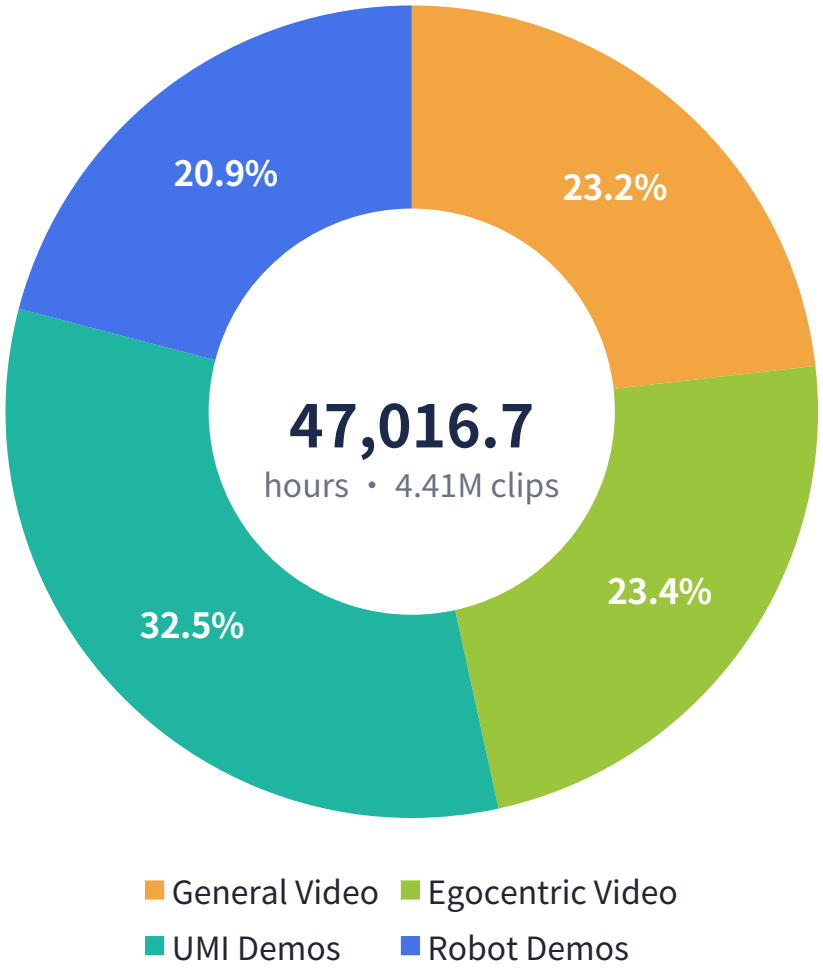
Scaling Up

Is more training better?

RoboTwin Hard: 0 h 34.56 → 67.5k h 59.60; real-robot Hard SR reaches 48.89 at 90k h

Data

47k h, 4 data types: UMI dominates with scalable action labels



General Video

10,909.6 h

Panda-70M、 OpenVid

Language–video: general visual dynamics

Egocentric Video

10,986.3 h

EgoSuite、 Ego4D、 EgoVerse

Language–video: human hand interaction

UMI Demos (in-house)

15,284.9 h

RoboPocket: phone + handheld adaptive gripper, 412k episodes

Lang.–video–action: scalable human manipulation

Robot Demos

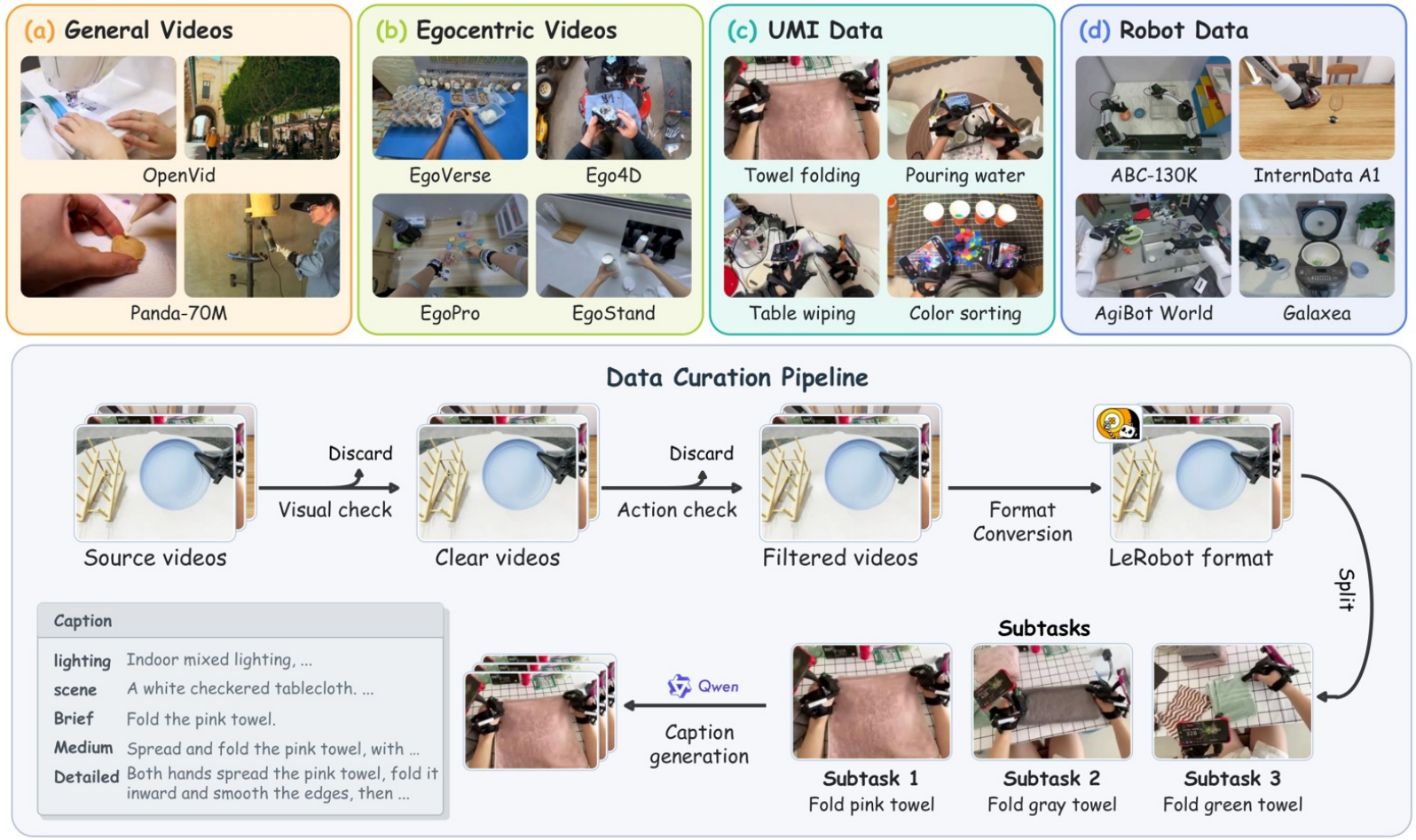
9,836.0 h

ABC、 AgiBot、 InternData-A1、 Galaxea、 RoboCOIN、 RoboMIND

Lang.–video–action: direct but limited robot data

Data Processing

QC → unify → segment → 3-level captions + scene/light



1 **Visual QC**

VLM filters animation, static, blurry, dark, occluded, and no-action clips

2 **Action QC**

NaN/invalid quaternions; joint |q|>20 rad, frame-to-frame jump >1 rad; EE shift >0.2 m or rotation >90°

3 **Unified Format**

LeRobot 2.1; 16-D bimanual EE (xyz + wxyz quaternion + gripper), per-source q1/q99 normalization

4 **Subtask Segmentation**

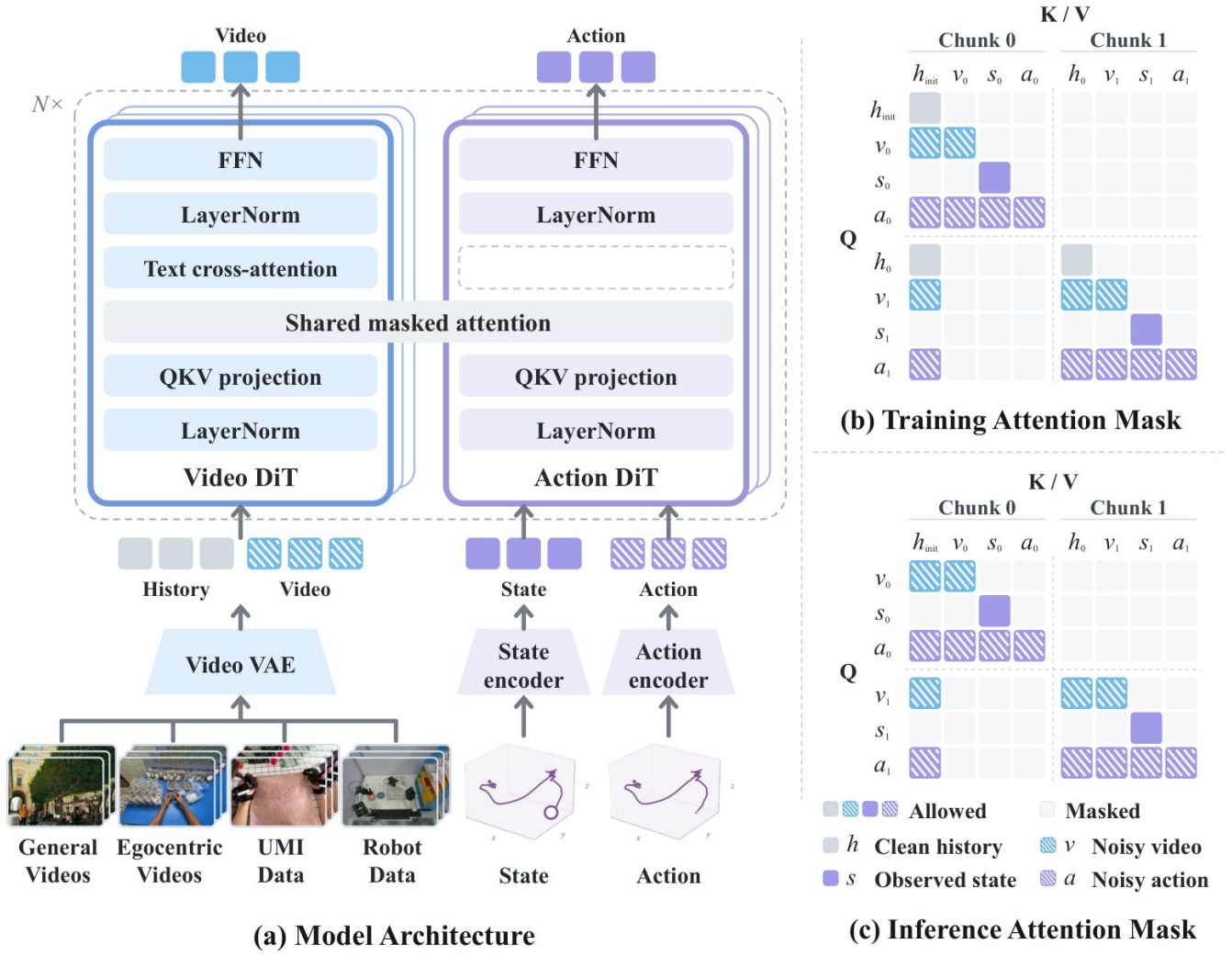
Qwen 3.8 Max splits by action intent into non-overlapping subtasks (each robot episode = one subtask)

5 **Caption**

Seed 2.1 Pro writes Brief / Medium / Detailed captions + scene + lighting

Model

Dual-stream MoT WAM: video→action only; no text to action



(a) Model Architecture

(b) Training Attention Mask

(c) Inference Attention Mask

Dual-Stream MoT

Video DiT initialized from Wan2.2-TI2V-5B; action DiT (state/action encoders, Transformer, decoder) randomly initialized.

One-Way Shared Masked Attention

Action tokens see past and current video; video tokens can’ t see actions → action-free videos train only the video branch.

Language Reaches Action via Video

Only the video branch has text cross-attention, reusing pretrained language–vision correspondence.

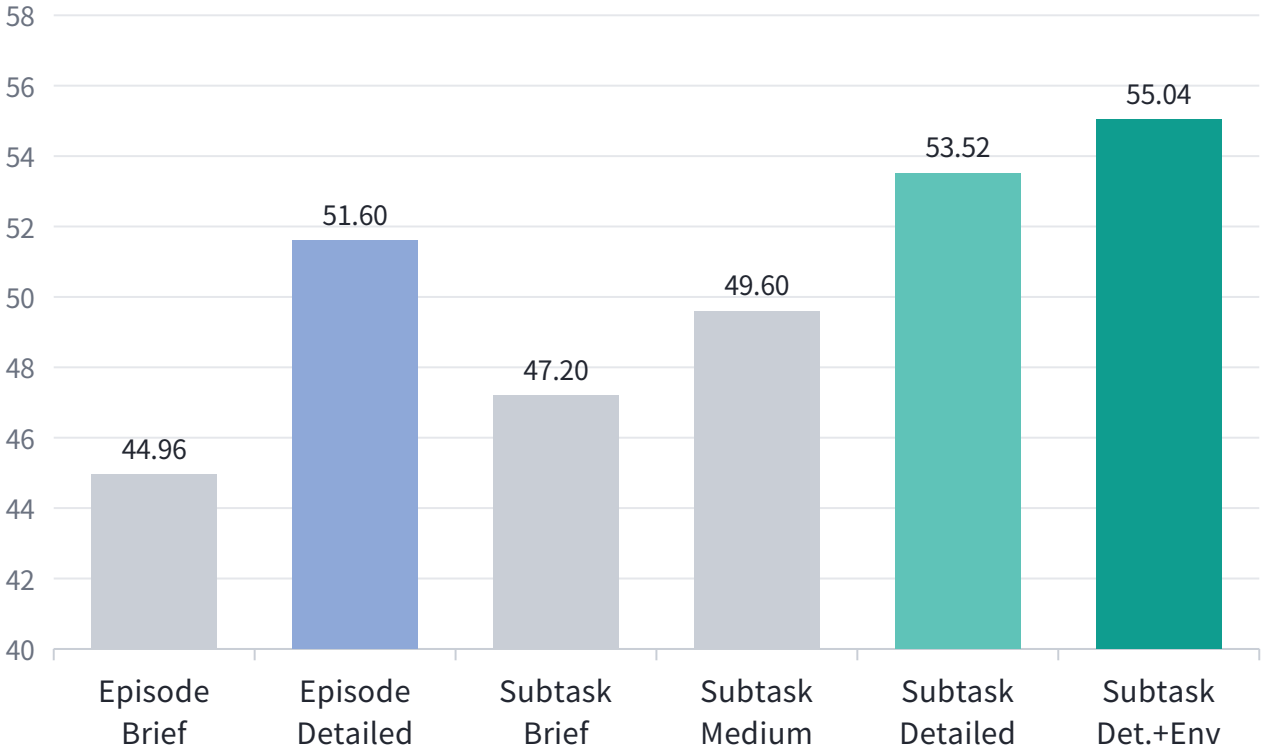
Joint Flow Matching

L = L_video + L_action (equal weights); each chunk: 2 video latent frames + 48 action steps; chunk-wise inference.

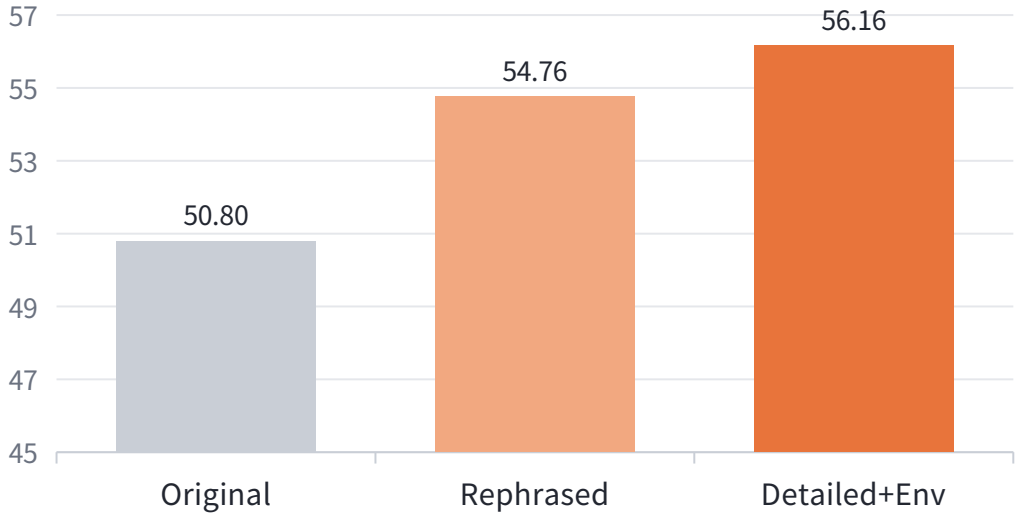
Finding 1 • Language–Video Alignment

Finer, environment-aware, subtask-aligned captions generalize better

Ego + UMI pretraining, 20k steps, different caption designs



Open-source robot data: rewritten captions



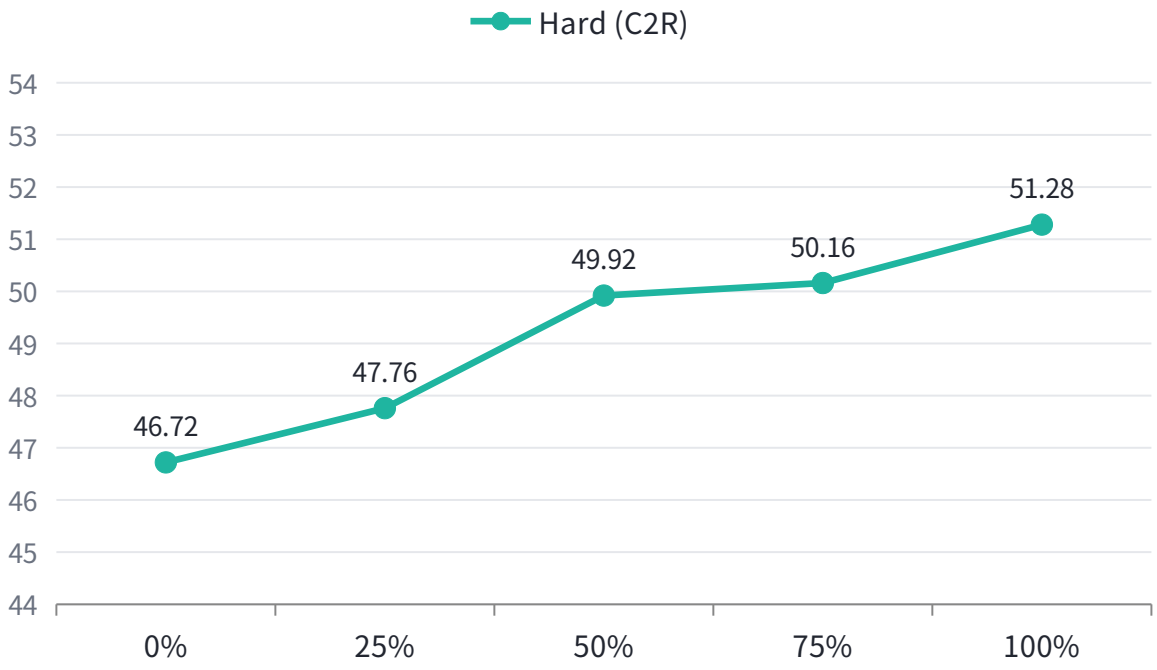
145,430 trajectories share only **592** task instructions
Just 100 paraphrases each give +3.96 pt

Accurate alignment matters too: shuffling Detailed subtask captions across clips within a trajectory drops Hard 53.52 → 52.00; **Easy differs by only ~1–2 pt across variants, so gains lie mainly in generalization.**

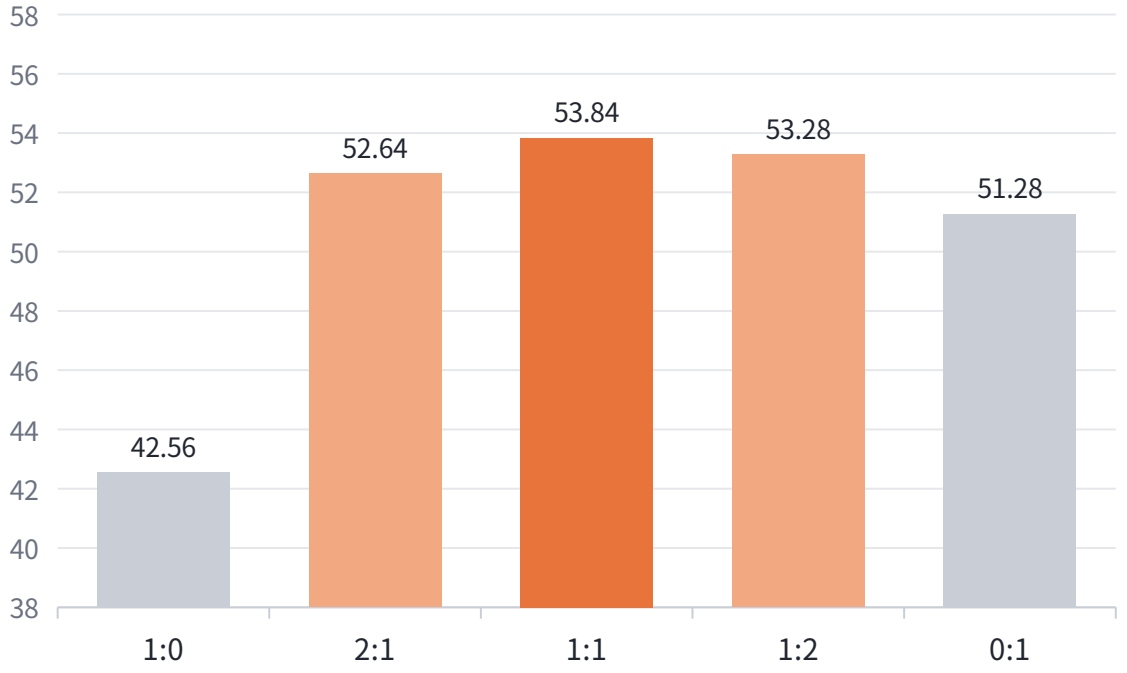
Finding 2 • Language–Video–Action Alignment

Action labels from UMI human demos help; a 1:1 Ego–UMI mix is best

UMI action-label coverage (all videos kept)



Ego : UMI sampling ratio (fixed 20k-step budget)



+4.56 pt

Action coverage 0%→100%: Hard SR rises steadily; UMI looks unlike robots, yet its action supervision transfers

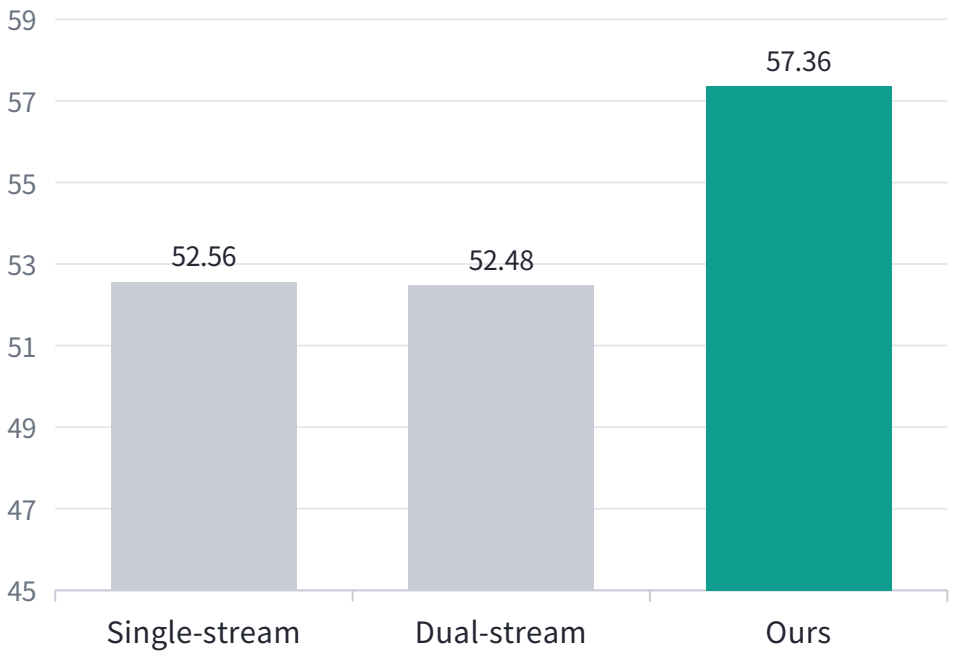
Mixed > Single Source

All three mix ratios beat Ego only (42.56) or UMI only (51.28): the two are complementary

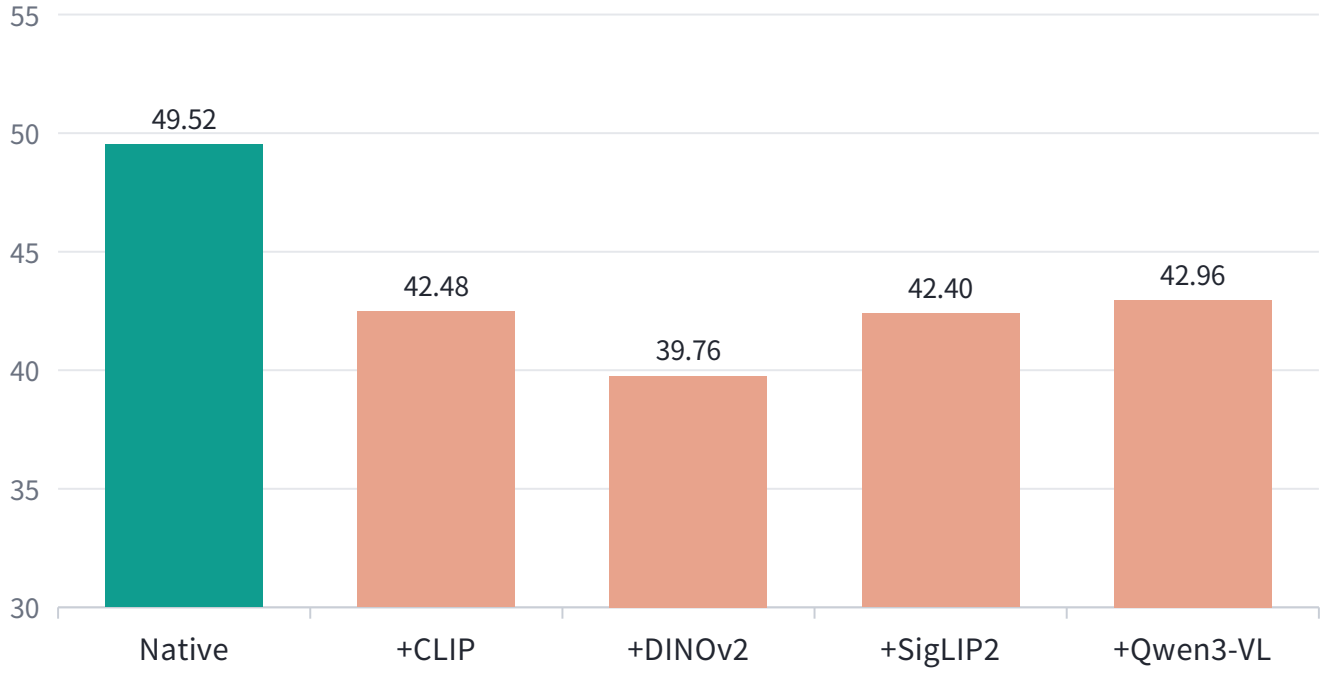
Finding 3 • Co-Training

Action reads text only via video; extra encoders hurt OOD

Arch./lang. path (30k-step multi-src PT)



Extra image features (injected via cross-attention)



+4.88 pt

Removing text cross-attn from action branch (language via video only): 52.48 → 57.36 vs. standard dual-stream

−6.6 ~ −9.8 pt

New visual path lacks large-scale language–video pretraining, so the model leans on in-distribution appearance cues

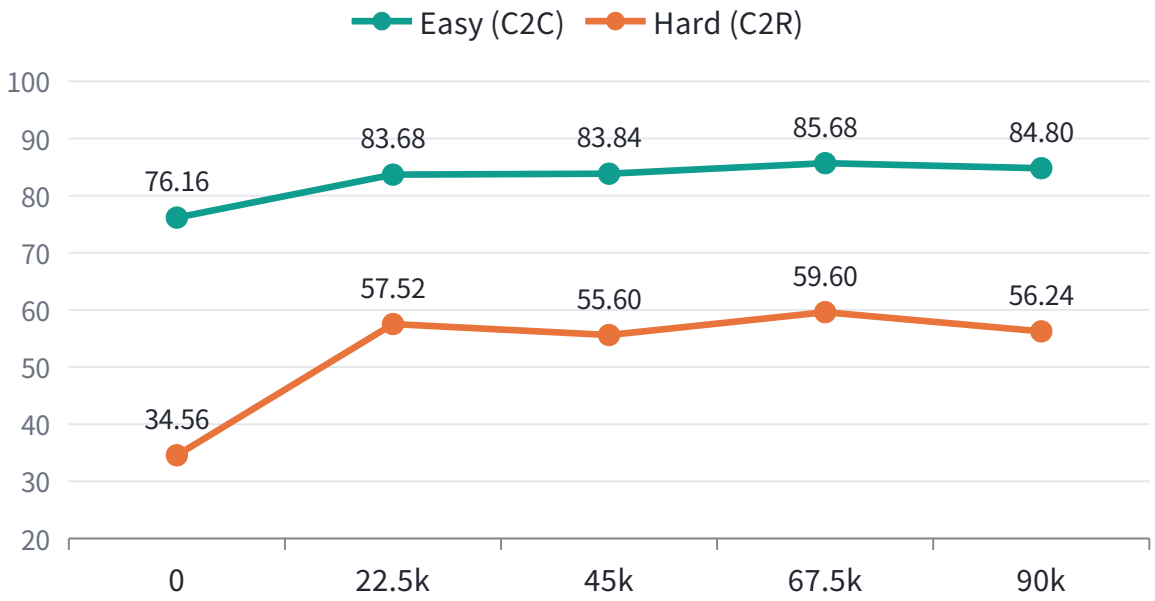
+2.64 pt

25k-step transitional pretraining on general video, then Ego+UMI mix: Hard 53.84 → 56.48

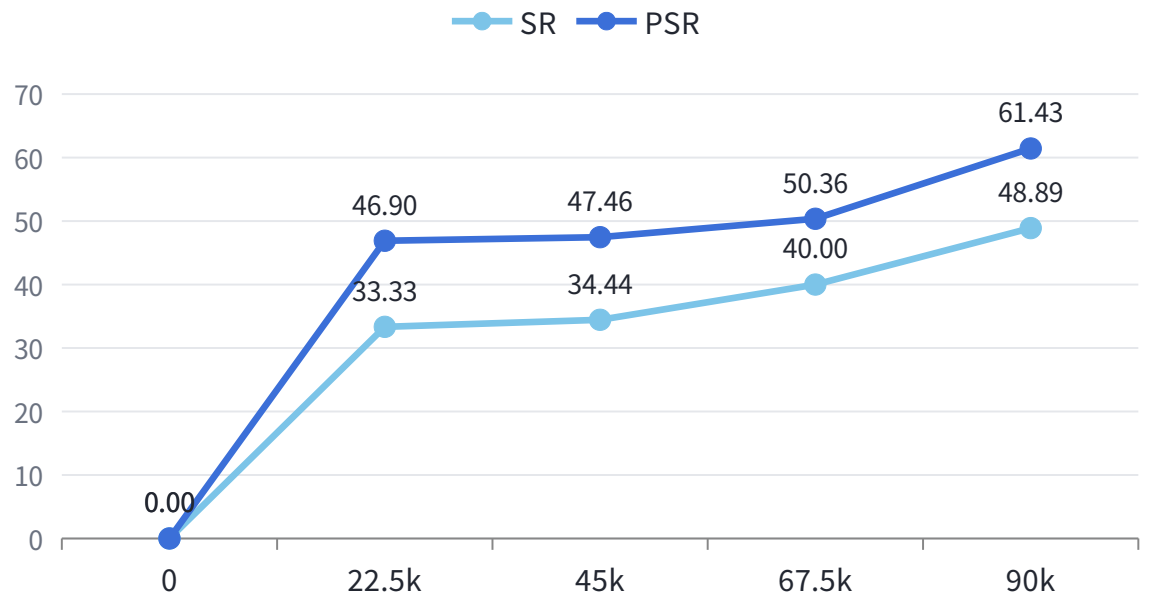
Finding 4 • Scaling

More pretraining helps greatly; sim is non-monotonic

RoboTwin SR (%) vs. training video hours



Real-robot Hard (avg. of 3 tasks, %)



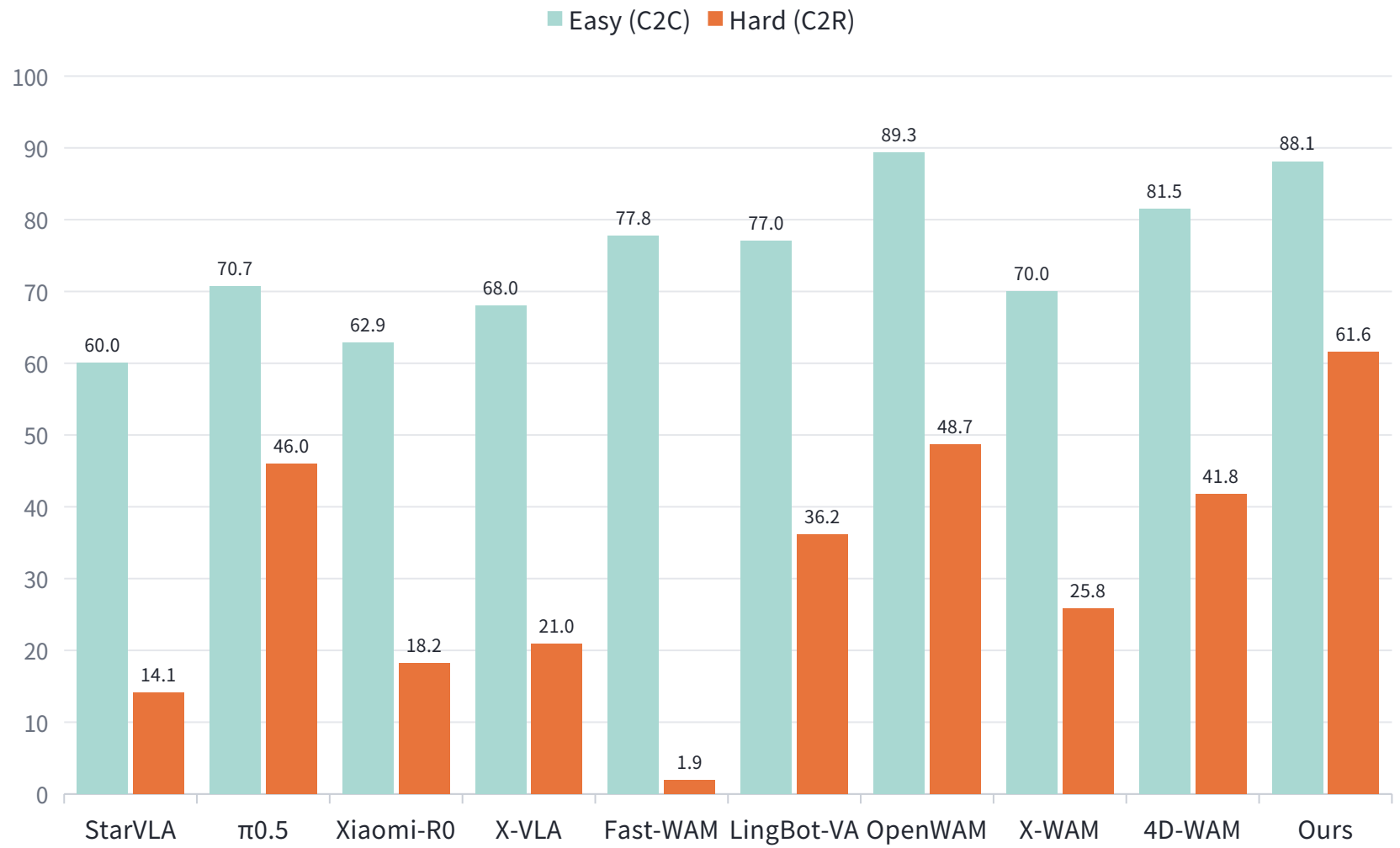
Two-stage: Ego+UMI (Stage 1) → robot data (Stage 2), RoboTwin Hard

34.56	53.84	59.60	61.52
0k / 0k	20k / 0k	75k / 0k	75k / 25k

Note

Sim Hard peaks at 67.5k h (59.60), then drops to 56.24 at 90k h;
real-robot Easy avg. SR 57.78 → 48.89, not rising with scale (Fig. 12). The 0 h model fails all real-robot trials.

RoboTwin 2.0: ≈OpenWAM on Clean, +12.86 pt Rand.



61.56%

Hard SR, best overall; OpenWAM 48.70%

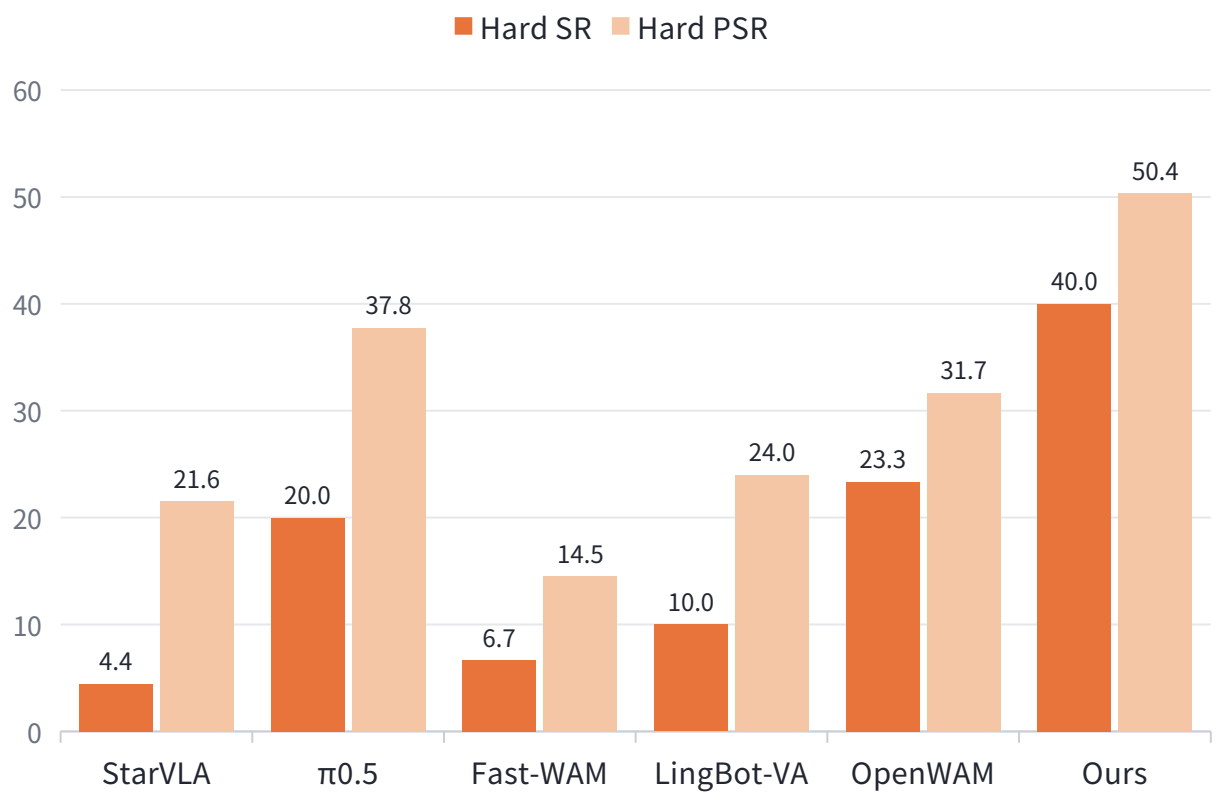
74.83%

Easy/Hard avg., best; OpenWAM 69.00%

77.76 → 1.92

Fast-WAM: high on Clean, near zero in randomized scenes; Clean scores don’ t reflect generalization

Real Hard: SR 40.00%, +16.67 pt vs. best baseline



Easy: Ours has the best SR (56.67); PSR 68.57, slightly below π0.5' s 71.87
Tasks: cup stacking (7 subtasks), towel folding (4), gift boxing (2)

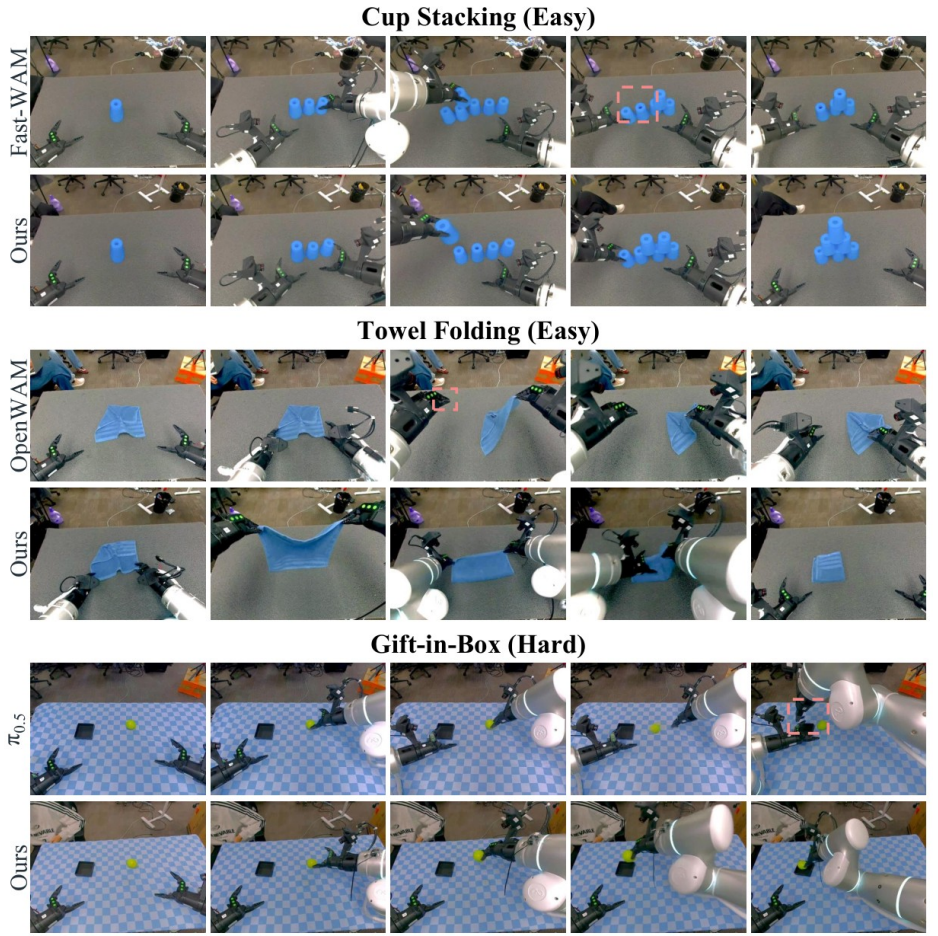
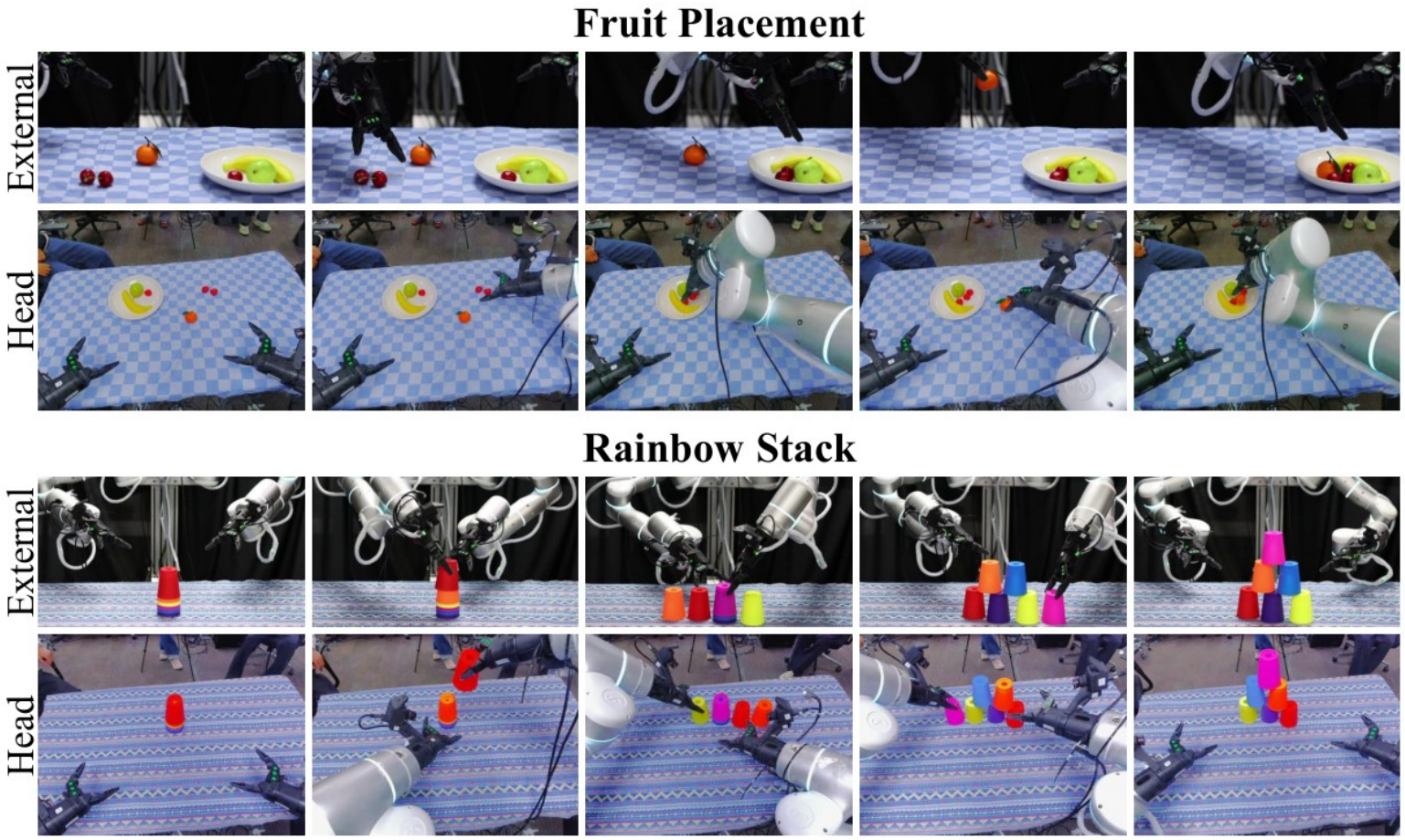


Fig. 7: baseline failures (red): cup collapse / one-sided towel lift / gripper slip

Qualitative Results

Skills transfer to new objects, containers and tabletops



Fruit Placement

Done by the gift-boxing model: scene, manipulated object and target container all differ from training.

Rainbow Stack

Trained only with same-color cups on a gray tablecloth, the cup-stacking model stacks multicolor cups on a patterned cloth.

Critical Assessment

Valuable; some scaling/real-robot claims need caution

Scaling Not Always Monotonic

Sim Hard peaks at 59.60 (67.5k h), falls to 56.24 (90k h); real-robot Easy avg. SR 57.78 → 48.89. Only real-robot Hard rises monotonically.

Checkpoint Trade-off

Table 11 real-robot 56.67 / 40.00 matches the 67.5k h point, not the Hard-best (48.89) 90k h point: Easy vs. Hard trade-off.

Real-Robot Gains Task-Specific

Fig. 11 (est.): towel Hard ~53% vs. baselines ≈0; cup Hard SR≈0 for all, Ours PSR below π0.5; small gap on gift boxing.

Limited Ablation Rigor

Only 25 episodes per task in ablations, no multi-seed variance; many gaps are 1–3 pt (e.g., caption shuffling only −1.52).

Exposure–Budget Confound

In Table 7, more data exposure also means more total steps; UMI 10k→20k even drops Hard 53.84 → 51.20.

Stated Limitations

Limited complex reasoning and explicit multi-stage planning; short history context limits long-horizon tasks.

Summary & Takeaways

Data & supervision matter as much as architecture

1

Treat captions as a core asset

Per-trajectory, subtask-level captions with scene/lighting clearly beat one shared task instruction.

2

Scale action supervision with low-cost human demos

UMI handheld-gripper action labels transfer to robots; a 1:1 mix with egocentric video works best.

3

Reuse the video backbone’ s pathways

Actions read language only via video; no extra CLIP/DINOv2-style encoders; adapt first on general video.

4

Evaluate generalization

With similar Clean scores, randomized scenes and real-robot Hard separate methods (61.56 vs 48.70; 40.00 vs 23.33).

Open Resources

- 47k h multi-source data (incl. 15k h in-house UMI)
- Training code
- Model weights

Future Work (authors)

- Memory mechanisms
- Hierarchical planning

Thank you! Questions welcome